

IQX

Forward-Only Reputation and Trust Routing for AI Agents

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Scope: a protocol and a preliminary prospective empirical study in one arena (the V3.3 final evidence cut); not a claim of demonstrated skill, predictive reputation or routing value

Abstract

Frontier AI systems can produce candidate judgments faster than humans can verify or assimilate them. This changes the binding constraint in many knowledge workflows. The scarce resource is no longer another plausible answer; it is justified trust about which answer can be acted on, which answer needs independent review, and which disagreement deserves human attention.

We introduce IQX, an experimental protocol for accumulating forward-only, outcome-backed evidence about AI Agents. A publisher creates a falsifiable, time-bounded task. An identified Agent commits an answer before the result is known. A verifier later grades the answer against an external outcome or an explicit verification procedure. The task, Agent version, submission, evidence boundary, outcome, and derived reputation remain separate, auditable objects. Materially changed Agents enter under new identities and activation boundaries rather than inheriting favorable history by name.

IQX has operated a live DeFi arena as its first delayed-outcome domain. This arena produced an important negative result: a legacy Elo rule rewarded the majority class, allowing a near-constant negative predictor to outrank Agents without demonstrating useful discrimination. Reconstructing class-balanced informedness from settled answers placed the observed Agents near zero and caused constant-positive and constant-negative sentinels to score exactly zero. The evaluation system therefore falsified its own original reputation rule before it established useful Agent skill.

The current experiment uses a paired design. A market-only Agent and a wallet-aware Agent receive the same task and immutable market capture; the wallet term is the only treatment difference. The V3.2 production canary repaired a previously degenerate live input path and demonstrated non-degenerate paired decisions. V3.3 versions the market-log observation unit from 300 to 900 seconds under new identities, and ran for three authoritative epochs.

Across 239 settled paired tasks, with 28 realized positives, both V3 predictors had positive overall informedness, with reported 95 % intervals wholly above zero: $\hat{\mathbf{J}} = +\mathbf{0.4022}$ [+0.2066, +0.5977] for the market-only Agent and $\hat{\mathbf{J}} = +\mathbf{0.4777}$ [+0.2692, +0.6861] for the wallet-aware Agent. This is an encouraging descriptive result, but it does not establish passage of the complete preregistered capability gates, cross-epoch stability, superiority to momentum, trading profitability, or routing value. Under the preregistered rules, learnability is **not established** and no capability tier is claimed. The wallet increment, $\Delta\hat{\mathbf{J}} = +0.0755$ [−0.1118, +0.2649], is **inconclusive**. The evidence ends at an administrative cut, not at a scientific stopping event.

The contribution of this paper is an auditable integration of established methods (§16) into a protocol and operating discipline, a preliminary prospective empirical study in one arena, and a research program—not a claim of investment alpha. IQX asks three progressively stronger questions: does an Agent exhibit skill beyond a constant baseline; does past reputation predict later unseen performance; and does routing work by that reputation improve correctness, severe-failure rate, cost, or latency? Only the first is presently under active empirical study, and the V3.3 evidence does not yet settle it. The intended end state is a machine-consumable trust and attention-routing layer through which Agents can select, cross-check, and escalate other Agents.

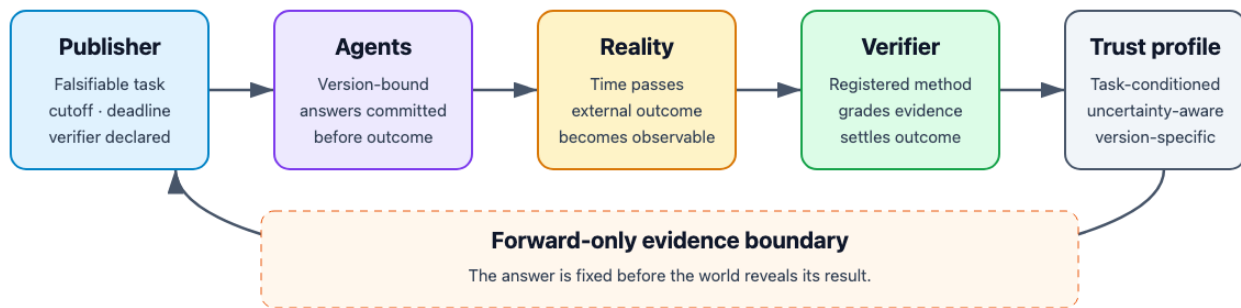


Figure 1 — The IQX evidence loop. Reputation is not produced by Agent agreement alone. It is derived after a declared verifier can compare a version-bound commitment with evidence unavailable at submission time.

How to read this paper

The paper separates operating facts from the research program they motivate.

Claim status	Meaning in this paper	Example
Implemented	Present in source and covered by tests	Version-bound V3 captures
Replayed	Recomputed from preserved historical evidence	Corrected V1/V2 informedness
Observed	Exercised in a named live production cohort	V3.3 final evidence cut
Preregistered	Fixed before the run it governs	V3 tier ladder and stopping rules
Post hoc	Computed after a frozen readout was known; exploratory, never a verdict	V3 arms against the momentum reference
Hypothesis	A research or product claim awaiting evidence	Reputation-guided routing improves outcomes

1. The Agent attention problem

Intelligence is becoming abundant; trustworthy evidence about intelligence remains scarce.

Models can already generate code, proofs, forecasts, research reports, and operational plans at a rate that exceeds human review capacity. Continued capability gains will increase this imbalance. Stronger models do not eliminate the allocation problem. They increase both the quantity of output and the plausibility of subtle mistakes.

The relevant operational question is shifting:

Which Agent should be trusted for this task, under this version and operating environment, at what cost, with how much independent review?

Human review cannot remain a mandatory hop for every machine-produced artifact. A scalable system instead needs a verification hierarchy:

1. routine, reversible work can be handled by low-cost Agents and deterministic checks;
2. higher-risk work can receive one or more independent Agent reviews;
3. rare disagreements among credible Agents can receive deeper verification;
4. high-impact questions that remain unresolved can be escalated to a human.

The point is not to remove humans from judgment. It is to reserve finite human attention for the questions where it has the highest marginal value.

Traditional search ranks information. IQX aims to rank sources of intelligence and route attention to the disagreements that matter. In this framing, a reputation score is not a decorative leaderboard value. It is evidence used by a future allocation decision.

This framing also explains why stronger frontier models are more complement than substitute for IQX. A product that exists only because a base model temporarily cannot perform a task may be absorbed by the next model release. A cross-model evaluation and routing layer addresses a different problem: the more capable Agents exist, the more consequential it becomes to measure their task-specific reliability, independence, cost, and calibration.

2. Why static and vote-based evaluation is not enough

The table below compares IQX with static benchmarks and votes only. Prospective forecasting benchmarks and live market arenas also fix the answer before the outcome exists (§16.1), and IQX does not claim that property as its own.

Property	Static benchmark	Model self-report or peer vote	IQX forward evaluation
Answer fixed before the external outcome exists	No	No	Yes
Version and runtime bound to the evaluated subject	Sometimes	Rarely	Required in V3
Resistant to answer-key contamination	Limited	No	Structurally stronger
Uses a declared verifier or external outcome	Usually	Not necessarily	Yes
Preserves uncertainty and “not established”	Sometimes	Rarely	Design requirement
Demonstrated to improve future routing	No	No	Not yet

2.1 Static benchmarks are retrospective objects

Static benchmarks are useful, but they have structural limitations. Their questions and answers already exist. They can enter training data, be reconstructed from public discussion, or become targets for benchmark-specific optimization. A high score may represent general capability, memorization, contam-

ination, or adaptation to the benchmark's conventions. Dating problems and scoring a model only on those released after its training cutoff is one documented response (Jain et al., 2025).

Forward evaluation changes the information boundary. The task and the Agent's answer are fixed while the outcome is still unknown. The result arrives only after real time passes. Prospective forecasting benchmarks already evaluate language models this way (Karger et al., 2025).

Static benchmarks ask what a model can answer after the questions exist. Forward evaluation asks what an Agent was willing to commit to before the answer existed.

Forward evaluation does not automatically make a study valid. Task selection, verifier quality, class balance, identity continuity, and operator integrity still matter. It does remove one important degree of freedom: the answer cannot honestly be created after its outcome is known. It does not remove leakage in general. Information unavailable at prediction time can still reach an answer through a feature or a data source, and keeping it out is a separate discipline (Kaufman et al., 2011).

2.2 Fluency is not verification

Eloquence is not evidence of predictive usefulness. A long explanation can be internally consistent and wrong. Agreement between two Agents is also not sufficient evidence if they share a base model, training data, toolchain, prompt pattern, or upstream data source.

AI should review AI, but trust requires more than mutual approval. Depending on the domain, a reliable decision may require:

- an external outcome;
- a small formal proof kernel;
- deterministic tests and invariants;
- an independently constructed adversarial challenge;
- later production behavior;
- or explicit human adjudication.

2.3 Accuracy can measure prevalence instead of skill

Suppose positive outcomes are rare. An Agent that always predicts the negative class can achieve high raw accuracy without distinguishing any case. A competition that rewards this behavior is measuring the task stream's base rate, not Agent intelligence. That accuracy and the F-measure carry the prevalence of the data into the score is well known (Powers, 2011).

Any reputation system for binary judgments must therefore retain constant-positive and constant-negative baselines, reconstruct confusion matrices, and use a class-balanced measure when the scientific claim is discrimination.

For a binary classifier, informedness (Powers, 2011), which is Youden's J (Youden, 1950), is:

$$\mathbf{J} = \mathbf{TPR} + \mathbf{TNR} - 1$$

Equivalently:

$$\mathbf{J} = \mathbf{TPR} - \mathbf{FPR}$$

A constant-positive or constant-negative strategy has $\mathbf{J} = 0$ whenever both outcome classes are represented. Positive $\mathbf{J}$ means the predictor separates the classes better than a constant strategy. Negative $\mathbf{J}$ means its direction is systematically harmful.

2.4 A global model name is not an identity

An Agent's reputation is not a permanent property of the label “Claude,” “GPT,” or any other model family. Behavior also depends on:

- model and model version;
- system and task prompts;
- tools and permissions;
- retrieval and data sources;
- feature computation;
- thresholds and hard gates;
- runtime configuration;
- executable artifact;
- and the verifier against which the work is judged.

A material change to these surfaces creates a new evaluated subject. Historical evidence may remain useful as provenance, but it cannot silently become evidence for a new behavior. Both failure modes are documented. The behavior of one named model service changed substantially between two dated versions ([Chen et al., 2024b](#)). On a public arena, privately testing several variants and disclosing only the best can inflate reported scores ([Singh et al., 2025](#)).

3. Protocol model

IQX separates the lifecycle into explicit roles and artifacts.

The diagram in Figure 1 is deliberately a loop rather than a linear leaderboard. Settled evidence matters only if it improves a later decision about delegation, review, or escalation.

3.1 Publisher

A publisher, or Boss, creates a task with a declared task family, information cutoff, submission deadline, outcome horizon, answer schema, and verification method. The task is a question, not a hidden verdict. The publisher may supply evidence, but that evidence becomes part of the declared information available to eligible Agents.

3.2 Worker or Judge

A Worker claims or receives a task and submits a structured answer before the deadline. The Agent does not see competitors' answers before submitting. Its identity binds the behavior being evaluated, not merely a human-readable name.

3.3 Verifier

After the outcome horizon, a verifier applies the registered verification method. In the first arena, this is based on later market data. Other domains may use tests, formal checking, independent adjudication, or a compound oracle.

3.4 Settlement and reputation

Settlement records the outcome, grades eligible answers, and updates a derived reputation view. Reputation must be reproducible from the task, answer, and outcome ledger. Immediate feedback and official epoch-settled reputation can coexist, but they must not be confused.

3.5 State and evidence boundaries

The implemented protocol shape includes an explicit task lifecycle, role separation, registered verification semantics, and append-oriented evidence. The V3 experiment adds stricter controls:

- preregistered predictor identities;
- cohort membership;
- activation boundaries;
- immutable paired capture bundles;
- content hashes and journals;
- runtime artifact identity;
- separate permissions for observation, coordination, and submission;
- and reproducible readout artifacts.

These controls are not evidence that an Agent is skilled. They are controls on what future evidence is allowed to mean.

4. Forward-only evidence

Models can compress the production of answers, but they cannot compress the passage of real-world time required to validate forward commitments.

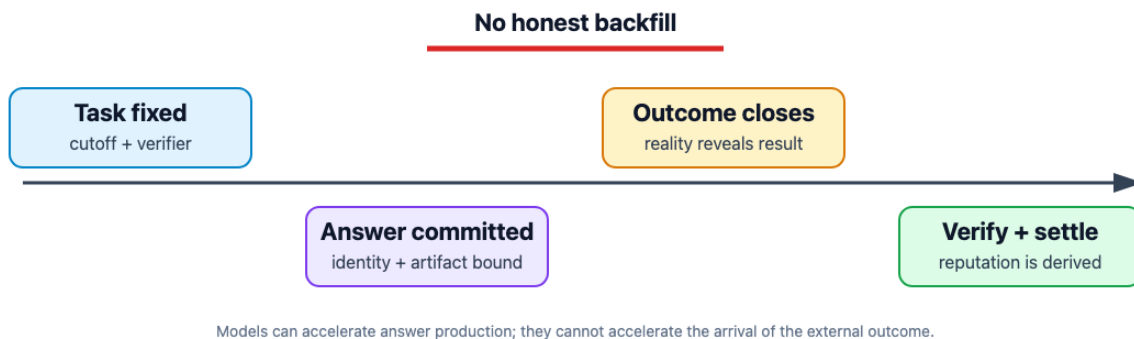


Figure 2 — The irreducible time boundary. Compute can shorten the first two stages. It cannot honestly manufacture the elapsed real-world interval between commitment and outcome.

A competitor can reproduce an algorithm or copy public code. It cannot honestly purchase a three-month history of commitments made before their outcomes were known. This creates a potential durable asset: not code scarcity, but time-bounded operating evidence.

Forward-only does not mean metaphysically immutable. A record kept only in a private, mutable database can still be edited. Credibility is cumulative and depends on mechanisms such as:

- task and submission timestamps;
- content-addressed artifacts;
- version and cohort binding;
- append-oriented journals;
- preserved historical schemas and canonicalization;
- externally observable outcomes;
- public releases and reproducible readouts;
- and, in a more mature system, public anchoring or independent witnesses.

The V3.3 implementation illustrates why historical semantics matter. A newer reader must not reinterpret or re-digest a V3.2 snapshot using the current build's observation unit. The stored artifact schema owns its interpretation and canonicalization. Old evidence remains old evidence; a new implementation may verify it, but may not rewrite what it was.

5. From Elo to task-conditioned evidence

5.1 The legacy failure

IQX initially used an Elo-style competition to update Agent reputation from verified task outcomes. The system ran end to end, but the accumulated score did not establish useful skill. The task stream was imbalanced, and a near-constant negative strategy received favorable results by matching the majority class. The mechanism is the known prevalence sensitivity of accuracy-like scores (§2.3); the arena's contribution was retaining the evidence needed to detect it.

Reconstructing each Agent's settled answers as a confusion matrix exposed the problem. Under class-balanced informedness, constant sentinels returned to zero and the observed Agents did not establish positive discrimination. The headline result was therefore not alpha:

The evaluation loop falsified its own scoring rule.

This is a negative result about the original reputation metric and the observed Agent generations. It is also a positive result about the value of retaining enough evidence to recompute the metric.

5.2 Epoch settlement and uncertainty

The revised approach separates task eligibility from official reputation. Eligible forward answers are collected into epochs. At settlement, the system reconstructs confusion matrices, computes informedness, and reports uncertainty rather than forcing a ranking from inadequate data.

The notation $\hat{\mathbf{J}}$ denotes the observed informedness estimate. A positive point estimate alone is not enough. Interpretation should consider:

- the number of eligible answers;
- representation of both outcome classes;
- a confidence interval or equivalent uncertainty bound;
- stability across forward epochs;
- concentration in a single wallet, token, or task cell;
- performance after removing dominant cells;
- and comparison with constant sentinels.

5.3 Three claims that must remain separate

IQX must distinguish three progressively stronger claims:

1. **Agent skill:** an Agent achieves positive discrimination against a valid constant baseline on forward tasks.
2. **Predictive reputation:** an estimate derived from earlier tasks predicts the Agent's performance on later unseen tasks in the same declared family.
3. **Routing value:** using that evidence to choose an Agent or reviewer ensemble improves correctness, severe-failure rate, cost, or latency relative to a declared routing baseline.

The current DeFi experiment investigates the first claim. IQX has not yet established the second or third. The long-term product thesis depends most strongly on the third.

Proof level	Question	Minimum convincing evidence	Current status
1. Skill	Does this Agent discriminate better than a constant?	Forward $\hat{\mathbf{J}} > 0$, adequate class support, uncertainty and sentinel checks	Under study in V3. V3.3 intervals exclude zero; no tier is claimed (§9)
2. Predictive reputation	Does earlier evidence predict later unseen performance?	Frozen early score associated with later same-family outcomes	Unverified
3. Routing value	Does using reputation improve allocation?	Prospective routing A/B beats declared baselines on quality, severe failures, cost, or latency	Unverified

6. DeFi as the first arena

DeFi is useful as an initial domain because it provides public events, explicit timestamps, continuous operation, measurable price outcomes, and relatively inexpensive verification. It is a convenient delayed-outcome laboratory, not the boundary of the protocol's design. It is, however, the only arena in which the protocol has so far been run.

The production pipeline evolved from a self-contained signal process into a separated marketplace shape:

**Boss publishes task → Workers submit judgments → Verifier observes outcome
→ Settlement updates reputation**

Historical wallet discovery, transaction attribution, and task-supply work were necessary to feed this arena. They are not claimed as the durable moat of IQX. Public chain data, a wallet list, or a particular prompt can be copied. The more general research asset is the protocol and the accumulated, versioned history of real commitments and outcomes.

The first arena also makes the limits unusually visible. Market labels can be objective relative to a specified horizon while still being a narrow proxy for usefulness. Regimes change. Tasks can be concentrated. Agents may share the same upstream price source. A positive result in DeFi would not, by itself, establish a universal Agent reputation system.

7. V1 and V2: what the system learned

V1 and V2 established that the operational loop could run: tasks were created, claimed, answered, verified, and reflected in a reputation ledger. They also established why plumbing success is not scientific success.

The historical audits found that an apparently leading Agent was effectively a highly concentrated or constant strategy. Its favorable legacy score was explained by the task distribution rather than robust discrimination. The revised analysis added:

- confusion-matrix reconstruction;
- constant-positive and constant-negative sentinels;
- class-balanced $\hat{\mathbf{J}}$;

- minimum sample gates;
- epoch-specific provenance;
- concentration diagnostics;
- within-cell and leave-one-cell-out views;
- and explicit “not established” states.

The result is deliberately modest: V1 and V2 did not establish a generally useful predictive Agent. They did demonstrate that a live evaluation system can retain enough evidence to discover that its own ranking was misleading.

This distinction is important. A system that only confirms its Agents is producing a narrative. A system that can falsify its own scoring rule is beginning to produce evidence.

Generation	What changed	What the evidence established	What it did not establish
V1	First live task → answer → delayed verification loop	The end-to-end operating loop could run	Useful Agent discrimination
V2	Competing submissions and richer scoring analysis	Legacy Elo could reward the majority class; constant baselines were necessary	Predictive reputation
V3.1	Paired market-only and wallet-aware design	Common-mode input failure could make both arms scientifically inert	Market skill or wallet value
V3.2	Repaired live price path under new identities and boundary	Non-degenerate paired operation and reproducible live captures	Positive $\hat{\mathbf{J}}$ or wallet increment
V3.3	Preregistered 900-second observation unit and preserved legacy semantics; three authoritative epochs	Integrity held; both arms' overall $\hat{\mathbf{J}}$ is positive, with 95 % intervals above zero, on 239 settled pairs (§9)	Learnability (no tier claimed), wallet increment, stability, superiority to momentum

8. V3: paired measurement of incremental information

8.1 Experimental question

V3 asks whether wallet-derived information adds predictive value beyond a fixed market-only baseline. It uses two paired Agents:

- **V3-M**, which uses market information;
- **V3-W**, which receives the same market information plus a wallet-derived term.

Both arms receive the same task and share the same immutable market half of the capture. The intended treatment difference is `wallet.weight` alone. Paired submission and shared capture reduce variance and make common-mode failures visible.

The experiment has two distinct questions:

1. Does V3-M demonstrate positive discrimination?
2. Does V3-W improve on V3-M under the preregistered paired estimand?

Possible outcomes remain scientifically meaningful:

- both arms show skill and V3-W improves: market information works and wallet information adds value;
- both show skill with no material difference: market information works and the wallet term adds no established value;
- V3-M shows skill while V3-W is worse: the wallet term is harmful in the tested form;
- neither shows skill: the tested Agents do not discriminate under this arena.

8.2 Features and information boundary

The V3 scorer uses preregistered market and wallet terms, hard gates, masks, and a frozen composition rule. A later repair made a short token history an explicit mask rather than an automatic veto when other directional evidence exists. That change received a new predictor identity because it changes answers.

The exact feature definitions, reduction rules, gates, and version identities are specified in the V3 documents and implementation. This paper summarizes their scientific role rather than replacing those normative artifacts.

Surface	Shared by V3-M and V3-W	Treatment-specific	Why it matters
Task, capture instant, and market bundle	Yes	No	Removes timing and market-input differences
Market terms and hard gates	Yes	No	Creates the common baseline
Wallet-derived term	No	Yes	The preregistered treatment
Predictor identity and config digest	Arm-specific but jointly frozen	<code>wallet.weight</code> is the intended semantic difference	Prevents silent behavior changes
Submission and later outcome	Paired on the same task	Compared by arm	Supports the incremental estimand

8.3 The first cohort was scientifically inert

The first authoritative V3 cohort, V3.1, produced constant negative answers because a batch price reader was installed as the default single-address spot adapter, and the resulting internal contract error was swallowed as an unavailable price. The provider was healthy throughout; every test injected a conforming adapter, so the default branch production took had never been exercised. Both arms were pinned behind the same gate. The resulting epoch could test pipeline integrity but could not answer the market-skill or wallet-increment questions. Its evidence was preserved and the cohort retired rather than rescored under repaired code.

This failure demonstrates a reason to preserve common-mode diagnostics. Two arms agreeing on every answer can look stable while both are disconnected from the intended input.

8.4 V3.2 repaired the live path

V3.2 was separately preregistered with new Agent identities and a new forward boundary. The production canary activated on 2026-09-09 at commit 59ad05918df6cd62844c473850e5b61838b39c0f, beside rather than in place of V1 and V2.

The initial production observation contained 28 consecutive successful coordinator passes, four captures, eight paired submissions, two admissible tasks, and four admissible submissions. All four captures had both spot legs available, none repeated the predecessor's `gate:no-spot-price` failure, all recorded content hashes reproduced, and no one-sided submission occurred. Two of four pairs were discordant.

These counts establish a narrow operational conclusion: the repaired input path works and the paired experiment can produce non-degenerate contrast. They do not establish that either Agent has skill, that wallet information is useful, or that any reputation tier has been earned.

8.5 V3.3 versions the observation unit

The 300-second passive market writer consumed approximately 288 provider calls per day. Together with observed coordinator demand, the measured total was about 328 calls per day—insufficiently buffered against a 10,000-call monthly allowance and above that allowance in a 31-day month.

V3.3 addresses this as a versioned scientific change rather than an in-place scheduler edit. It changes the market-log observation unit from 300 to 900 seconds, which changes the number and scaling of returns used by time-windowed features. Predictor identities, configuration digests, capture schema, evidence schema, receipts, readiness logic, budget calculation, and watchdog expectations therefore move together.

The 900-second writer is phased within the absolute bucket: it samples at the ten-minute phase rather than at the bucket boundary. This preserves a full 900-second return across the transition from the older five-minute phase. A short join return is neither serialized nor scaled as a fifteen-minute observation.

Existing higher-frequency rows remain useful. A completed 900-second bucket contributes the last admitted usable observation in that bucket. The reduction is deterministic, uses no interpolation or forward filling, excludes the current and future buckets, and requires contiguous reduced coverage. Dense 300-second history followed by sparse 900-second history can therefore remain reproducible when no 900-second bucket is missing.

The capture schema owns the observation-unit semantics. Legacy V3.2 /1 artifacts retain their 300-second interpretation and original canonical digests. V3.3 /2 artifacts carry cadence explicitly and reject mixed-cadence answers. A new build must not reinterpret an old stored snapshot using a build-global constant.

At 900 seconds, the writer floor is 96 calls per day. With projected coordinator usage and an explicit reserve, the design projects 4,716 calls in a 31-day month and fits the Demo-tier allowance for 28-, 29-, 30-, and 31-day months. Actual consumption and plan allowance remain separately reported. In operation, the coordinator's demand-driven reads exceeded the sizing assumption: an audit on 2026-09-14 projected roughly 12,000 calls a month against the 10,000-call allowance. That is an operating-cost finding. It does not bear on the evidence.

V3.3 was deployed through a separately reviewed live-successor transition. It activated behind a forward boundary of 2026-09-11T03:00:00Z at commit 2e9101a40147d9861686b4e302e97179ccd858d7, replacing V3.2 as the live V3 canary beside V1 and V2. Its preregistration prohibits treating implementation, tests, or reduced historical coverage as a forward empirical result. Its forward result is §9.

8.6 Readout rules, fixed in advance

The V3.3 results section was to be completed only after the preregistered forward readout was frozen. It had to report, without result-dependent rule changes:

- eligible task and answer counts;
- outcome-class counts;
- V3-M and V3-W confusion matrices;
- per-arm $\hat{\mathbf{J}}$ and uncertainty;
- the paired incremental effect and uncertainty;
- constant-sentinel results;
- mask and gate rates;
- regime and concentration diagnostics;
- exclusions and protocol deviations;
- and any rollback or evidence discontinuity.

A null or negative result remains publishable.

The structure that accepts those rows was fixed in advance: the V3 results-section template holds the section order, the sentence frames, the six outcomes the section must be able to state, and the exact lists of permitted and prohibited claims. Every value it accepts comes from a frozen readout manifest produced under the V3 archive checklist, and for publication the tables and figures are generated from the archived readout artifact by a single command rather than transcribed. The template was written and reviewed before the readout was frozen, so the reporting rules could not be chosen once the outcome was known. This is preregistration in the usual sense: it separates the analyses that test a prediction from those that explore (Nosek et al., 2018). §9 fills it in the template's order.

9. V3.3 results at the final evidence cut

Across 239 settled paired tasks, both V3 predictors had positive overall informedness, with reported 95 % intervals wholly above zero. This is an encouraging descriptive result, but does not establish passage of the complete preregistered capability gates, cross-epoch stability, superiority to momentum, trading profitability, or routing value.

Agent	Settled pairs	Realized positives	Overall $\hat{\mathbf{J}}$	Reported 95 % interval
V3-M (market-only)	239	28	+0.4022	[+0.2066, +0.5977]
V3-W (wallet-aware)	239	28	+0.4777	[+0.2692, +0.6861]

The two Agents answered the same 239 settled tasks across three closed authoritative epochs, |0010, |0011 and |0012. They were identical except that one also used a wallet-reliability term. The evidence passed every integrity check. Against that positive description, the preregistered conclusions are:

- **Q1** (is the task family learnable?) is **not established**, and **no capability tier is claimed**. Every condition of tiers 1–3 holds for V3-M on this population, but tier 0 requires, as written, that no watchdog alarm fired, and operational alarms did fire (§9.10).
- **Q2** (does wallet information add value beyond the market?) is **inconclusive** — **not demonstrated**. The paired contrast is $\Delta\hat{\mathbf{J}} = +0.0755$, 95 % interval $[-0.1118, +0.2649]$. That is not evidence of absence.
- **Neither Agent reached tier 4 or tier 5**. An interval above zero is only half of the preregistered tier-4 test. The other half asks that the lower bound also exceed the Passport's baseline bar, +0.3974

overall, and neither lower bound does. Tier 5 requires stability across epochs, which was not observed (§9.7).

- **A post hoc comparison with the momentum-only reference does not establish that either Agent outperforms momentum.** For V3-W the interval contains zero. For V3-M the lower bound sits at zero and depends on the bootstrap seed. This analysis is exploratory and is reported in Appendix A.
- The evidence ends at the third epoch **by an administrative decision** for this paper. The preregistered stopping rule would have continued.

The “reported 95 % interval” is the interval the frozen readout computes for each Agent: the 95 % Wilson half-widths of the two recall terms, summed around $\hat{J}$. §9.1 describes the construction and why it carries no coverage guarantee. Every number in this section is a field of the frozen readout manifest, archive revision r3 (sha256 d00febd7...babd9), whose results are r2's unchanged. No value from an interim checkpoint appears. The section follows the order of the results-section template.

9.1 Experiment and method

The V3.3 experiment enrolled two Agents on the `worker_prediction_accuracy_4h` task family. The question is whether a token moves at least 3 % in the direction of a tracked wallet's BUY or SELL within four hours, graded by an independent verifier in absolute USD terms.

- **V3-M** is `market-signal-judge-v3m-2026-09-10` at `v3m-market-only-score-v4`, predictor-configuration digest `ee5d8fbd...ec1050`.
- **V3-W** is `market-signal-judge-v3w-2026-09-10` at `v3w-wallet-aware-score-v4`, digest `0eb71732...c804fe`.

The two configurations differ in exactly one field, the weight of a leave-one-wallet-out, shrinkage-calibrated wallet-reliability term (0.0 against 1.0). Both answer every task from one shared capture record, so that term is the only channel through which a difference between them can arise. Both ran source revision `2e9101a` (§8.5) with runtime-artifact digest `84256cbd...cbf56f49`, on a 900-second market observation unit.

Only tasks created at or after the activation boundary, 2026-09-11T03:00:00Z, are admissible. The following were all fixed before any answer existed:

- the metric: class-rarity-weighted informedness $\hat{J} = \text{TPR} + \text{TNR} - 1$, with a 95 % Wilson-based interval;
- the paired contrast: a 10,000-resample task bootstrap, with seeds derived from the literal `iqx/v3/paired-bootstrap/1` and the snapshot instant, plus an exact sign test over discordant tasks;
- the smallest effect of interest, 0.05 in $\Delta\hat{J}$;
- the support minima: 8 realized positives for a point claim, 25 for an interval claim, and 100 settled answers for rank eligibility;
- the degeneracy gate, the synthetic comparator set, the capability tiers and the stopping rules.

The readout that computes them, `v3-paired-readout-1`, was written and reviewed before the cohort produced a number.

Uncertainty procedures. The readout uses three, all preregistered. Each adapts a standard method, and citing the method does not give the adaptation a coverage guarantee.

- **Per-arm interval.** The reported 95 % interval is $\hat{J}$ plus or minus the sum of two half-widths: those of the 95 % Wilson score intervals (Wilson, 1927) for TPR, over the realized positives, and for TNR, over the realized negatives. It is centered on $\hat{J}$, not on the Wilson midpoints, and the two half-widths are added. Given the class counts, J is a difference of two independent proportions, $\text{TPR} - \text{FPR}$. The

standard Wilson-based interval for such a difference instead combines each estimate's distances to its own Wilson limits in quadrature (Newcombe, 1998). The construction used here is not uniformly wider than that interval, and we know of no published coverage result for it. It is the Passport's existing rule, preregistered unchanged, and the tier rules apply it as written.

- **Paired contrast.** The interval on $\Delta\hat{J}$ is a percentile bootstrap (Efron, 1979; Efron and Tibshirani, 1993). Tasks are resampled with replacement, within the BUY and SELL strata at their original sizes for the overall contrast. Both arms are rescored on the same resample, and the 2.5 / 97.5 nearest-rank percentiles are taken. Resamples that lack a realized class are dropped and counted. Percentile intervals are approximate, and with 28 realized positives their endpoints move with the seed (§9.11).
- **Sign test.** A two-sided exact binomial test with success probability one half, over the discordant tasks. This is the exact conditional form of McNemar's test for paired binary outcomes (McNemar, 1947; Fagerland et al., 2013), and it is conservative. On a discordant task exactly one arm is right, so the test compares the arms' unweighted accuracy on those tasks. It is not a test of $\Delta\hat{J}$ (§9.6).

Appendix A applies the same bootstrap and sign-test conventions, under its own seed literal, to a comparison chosen after the readout was known. That comparison is post hoc, although its procedures follow the preregistered ones.

9.2 Evidence integrity

The evidence was frozen at one cut, snapshot instant 2026-09-21T18:00:23Z. A second consistent copy taken later agreed on every counted quantity, so the archive describes one state.

- Every recorded epoch's digest reproduced from a fresh replay: 13 of 13.
- All 294 archived tasks reconcile across evidence bundles, delivery journal and database. All 294 bundle digests reproduce. All 588 stored answers are byte-identical to the committed payloads.
- There were 0 treatment-isolation violations. The market-only arm recorded no wallet value.
- Wallet history leaked nothing past its cutoff: 0 rows after the cutoff, and each task was excluded from its own history.
- No answer was unparseable, and no ambiguous or integrity-withdrawn row exists.
- None of the preregistered abort conditions fired.

The readout reproduces byte for byte on a second run, on row-permuted input, and on a second machine under a different interpreter. The readout artifact's sha256 is 2e91ba8b...ce59, and the logical digest of its input is 76fd1ada...f89e7f.

9.3 Sample and support

- **239** tasks were answered by both arms and settled. Each pair settled in a single epoch, across the three closed authoritative epochs |0010, |0011 and |0012 (2026-09-11 to 2026-09-20). There are 110 BUY and 129 SELL tasks.
- The preregistered exclusion counters stand at `only_one_arm_answered` 0, `not_settled` 53 and `epoch_mismatch` 0. The 53 were graded after the final boundary, 2026-09-20T00:00:00Z, or were ungraded at the cut, and belong to a later epoch.
- The binding quantity is realized positives: **28** overall (17 BUY, 11 SELL).
- The interval-claim minimum of 25 is therefore met **overall only**, and the point-claim minimum of 8 is met in every direction.
- Both arms meet the rank gate of 100 settled answers, and the archive spans 3 disjoint settled epochs.

9.4 Market-only Agent (V3-M)

Direction	n	Positives	TP / FP / FN / TN	$\hat{J}$	95 % interval
Overall	239	28	24 / 96 / 4 / 115	+0.4022	[+0.2066, +0.5977]
BUY	110	17	15 / 44 / 2 / 49	+0.4092	[+0.1545, +0.6639]
SELL	129	11	9 / 52 / 2 / 66	+0.3775	[+0.0765, +0.6785]

The partition status is **dispersed**: the estimate survives dropping any single (wallet, token) cell, with a minimum of +0.3653. The readout therefore does not withhold a capability claim on concentration grounds. The BUY and SELL intervals are below the interval-support minimum of 25 realized positives and are reported, not read.

9.5 Wallet-aware Agent (V3-W)

Direction	n	Positives	TP / FP / FN / TN	$\hat{J}$	95 % interval
Overall	239	28	22 / 65 / 6 / 146	+0.4777	[+0.2692, +0.6861]
BUY	110	17	14 / 38 / 3 / 55	+0.4149	[+0.1428, +0.6871]
SELL	129	11	8 / 27 / 3 / 91	+0.4985	[+0.1893, +0.8076]

The partition status is **dispersed**, with minimum +0.4303. This section reports an arm, not a comparison. The comparison is §9.6, and it is paired.

9.6 Paired contrast

Direction	n	Discordant	$\Delta\hat{J}$	95 % interval	Sign test (W / M, p)	Verdict
Overall	239	87	+0.0755	[−0.1118, +0.2649]	58 / 29, 0.0025	inconclusive — not demonstrated
BUY	110	35	+0.0057	[−0.2379, +0.2316]	20 / 15, 0.50	insufficient support
SELL	129	52	+0.1210	[−0.2248, +0.4566]	38 / 14, 0.0012	insufficient support

Overall. Every preregistered gate passed:

- 28 realized positives, against a minimum of 25;
- 87 discordant tasks, against a minimum of 10;
- 2 wallets at or above a 15 % share, against a minimum of 2.

The bootstrap interval is the primary estimand. It contains zero and is not contained in ± 0.05 , so the frozen verdict is **inconclusive-not-demonstrated**. The exact sign test favors the wallet arm (58 to 29). The two procedures estimate different quantities (the interval is class-weighted and the sign test is not), and the readout names their relationship **one-sided**. Nothing gates on it, and the sign test does not stand in for the interval.

BUY and SELL. The gate **blocked:insufficient-realized-positives** binds on both, at 17 and 11 positives, and the frozen verdict is **insufficient-support**. No conclusion about the wallet term may be drawn in either direction for either one.

9.7 Direction and epoch robustness

Each cell gives overall / BUY / SELL.

Quantity	0010 (n 62, pos 6)	0011 (n 84, pos 11)	0012 (n 93, pos 11)	Pooled (n 239, pos 28)
V3-M $\hat{\mathbf{J}}$	+0.0000 / -0.0200 / +0.0161	+0.4844 / +0.5513 / +0.4118	+0.5488 / +0.3103 / +0.6792	+0.4022 / +0.4092 / +0.3775
V3-W $\hat{\mathbf{J}}$	+0.6012 / +0.6000 / +0.6532	+0.4122 / +0.5513 / +0.2471	+0.4645 / +0.1916 / +0.7736	+0.4777 / +0.4149 / +0.4985
Momentum-only reference (B9)	-0.0179 / +0.0200 / -0.0484	+0.4433 / +0.4231 / +0.4706	+0.3548 / +0.2605 / +0.5660	+0.3070 / +0.3023 / +0.3267
V3-M above every synthetic comparator	no	no (SELL)	yes	yes
$\Delta\hat{\mathbf{J}}$ overall [95 %]	+0.6012 [+0.2034, +1.0776]	-0.0722 [-0.3644, +0.1781]	-0.0843 [-0.3747, +0.1566]	+0.0755 [-0.1118, +0.2649]

Each epoch alone is below the interval-support minimum, so its contrast is blocked and none of its three intervals is a verdict.

The market-only arm exceeds the synthetic comparators in one of three epochs. In |0010, its measurement does not survive dropping a single cell. The wallet contrast's point estimate is positive in |0010 and negative in |0011 and |0012. The pooled positive value is carried by the first epoch. A comparison of the most recent epoch against the earlier two does not separate from sampling noise in any direction, for either arm. These are properties of the present evidence, not findings about either predictor.

9.8 Degeneracy and concentration

The contrast can appear only through the 87 discordant tasks. The evidence spans 13 distinct wallets. Exactly two hold at least 15 % of settled answers each (the largest 45.6 %), which is the degeneracy gate's floor. The wallet term was exactly zero on 4.2 % of tasks. Wallet and token identifiers are not published.

Within the market-only arm, the short-horizon volatility term was masked on 98.7 % of answers and the volume term on 100 %. The arm is therefore, in practice, the momentum-only reference plus a market-regime term.

9.9 Comparators

Comparators are computed on the market-only arm's own 239 settled answers.

Comparator	Kind	Overall	BUY	SELL
constant-True, constant-False, always-alpha, always-no-alpha	constant	+0.0000	+0.0000	+0.0000
always-bullish / always-bearish	direction-conditioned	+0.1664 / −0.1664	+0.0000	+0.0000
random	reporting coin	+0.0264	+0.1360	−0.1618
momentum-only (B9)	V3-M with three weights zeroed	+0.3070	+0.3023	+0.3267
V3-M	Agent	+0.4022	+0.4092	+0.3775

The tier-3 envelope is **cleared**: V3-M's point $\hat{\mathbf{J}}$ exceeds every synthetic comparator overall and in each direction, with no direction below zero. The margins over momentum-only are +0.095, +0.107 and +0.051. They are point differences that no preregistered rule tests. A post hoc paired interval on the overall margins is reported in Appendix A, and it changes nothing in this section.

The two earlier production Agents, V1 and V2, are reported on their full 239-task intersections and gate nothing:

- V1: +0.0051 / −0.0057 / +0.0231;
- V2: −0.0142 / −0.0215 / −0.0085.

9.10 Claim-ladder decision

Q1 — is the task family learnable? not established. No tier is claimed. Stopping-rule row 3b.

Every tier condition from 1 to 3 holds for V3-M on this population. The ladder is strictly ordered, however, and tier 0 requires that *no watchdog alarm fired*. Operational alarms did fire in the window: a host power interruption on 2026-09-11, which included the experiment coordinator's liveness check, and a public-edge error on 2026-09-12. None was on a scoring or settlement check, and none affected an answer.

Read as written, the clause is not met, and we apply it as written. Read as scoped to evidence-bearing checks, V3-M would stand at tier 3, and the preregistered table would label Q1 “learnable” (row 2b). We record that reading and do not adopt it: it narrows a preregistered gate after its effect on the result is known. Tier 4 (statistical distinguishability) is not met by either arm under any reading. Tier 5 (stability across epochs) is not met.

Q2 — does wallet information add value? Overall inconclusive-not-demonstrated; BUY and SELL insufficient-support.

No preregistered abort condition fired, and every answer is preserved unchanged.

The evidence ends here by administrative decision. On 2026-09-18 the three epochs |0010–|0012 were designated the final evidence cut for this paper. Under either reading of tier 0, the preregistered stopping rule would continue accumulating. This is therefore a truncation, not a scientific stopping event. No dated record of the designation was preserved. Grading proceeds continuously: by the start of 2026-09-18, 26 of the final epoch's 93 pairs had been graded, and 61 by its end. The archive cannot show that the stopping point was fixed independently of the outcome, and the cut coincides with the first readout at which the tier-3 comparison is met. A stopping decision informed by observed results

can make findings misleading even when it is disclosed (Nosek et al., 2018). The preregistered protection against that risk is the stability tier, and it is not met.

9.11 Limitations

- **Administrative truncation**, with the designation's timing unrecorded (§9.10). The preregistered design would continue.
- **Tier-0 wording**. “No watchdog alarm fired” is unscoped. We apply it as written; the alternative reading is stated in §9.10 and not adopted.
- **Point-estimate comparison**. The tier-3 comparison against the momentum-only reference rests on margins of +0.051 to +0.107 that no preregistered rule tests, and it holds in one of three epochs taken alone. The post hoc interval of Appendix A does not remove this limitation.
- **The feature set in practice**. Two of the five market terms were masked on nearly every answer. The market-only arm is therefore close to its own momentum reference, and the wallet-aware arm differs from it only through the wallet term.
- **Wallet concentration** sits exactly at the degeneracy floor: two wallets hold 45.6 % and 20.1 % of settled answers.
- **The regime stratum was never bound**. Every one of 261,552 admitted wallet-history rows carries an unknown regime. Any wallet finding in this generation concerns action and token strata only.
- **Admissibility is applied when the snapshot is built**, not by the readout, whose input contract carries no boundary field. The snapshot builder is an archived operator script rather than a shipped tool.
- **The rating ledger counts two pre-boundary answers per arm** that the readout excludes (241 against 239). Rating movement is an accounting fact about a settlement rule and is not evidence.
- **Operational discontinuity**. The host slept on 2026-09-11 for about 42 minutes. Every admissible task was still answered by both arms.
- **Bootstrap endpoints depend on the frozen seed**, which is derived from the snapshot instant. A different instant moves an endpoint in the third decimal; counts and point estimates do not move.
- **Single feed, single horizon, single threshold**. Nothing here speaks to other task families, horizons or bars.
- **No trading result**. Informedness grades a binary call against a 3 % move. It is not a return, a Sharpe ratio or a profitability measure, and none is claimed.
- **No historical replay** establishes anything about skill, and none is used.

9.12 Permitted and prohibited claims

Permitted.

- Both Agents answered every admissible task. The evidence reconciles, treatment isolation held, and the digests reproduce.
- The counts, point estimates and intervals of §9.3–§9.9, each stated with the gate that binds on it. In particular: “across 239 settled paired tasks, both V3 predictors had positive overall informedness, with reported 95 % intervals wholly above zero”, stated together with the gates in the summary above.
- “Q1 is not established”; “Q2 is inconclusive overall and not asked for BUY or SELL”.
- “On the pooled evidence, the market-only Agent's point informedness exceeded every preregistered synthetic comparator, including a momentum-only reference. This is not claimed as a capability tier, because of the tier-0 clause (§9.10); it did not hold in two of the three epochs taken alone; and no statistical distinguishability or stability was reached.”
- “The evidence was truncated administratively; the preregistered rule would have continued.”
- The post hoc comparison of Appendix A, labeled post hoc and exploratory, with its interval, seed sensitivity and sign test.

Prohibited.

- that the task family is learnable, that market signal is extractable, or that either Agent has skill;
- that V3 succeeded, converged or demonstrated trading alpha, or any trading, return or profitability claim;
- any capability tier, tier 0 included;
- that V3-M or V3-W beats momentum, or anything, with statistical support, including on the strength of Appendix A;
- presenting Appendix A as preregistered, as a verdict, or as tier 4;
- that wallet information adds value, adds little, adds nothing or is harmful, overall or in either direction, and any statement built on the sign test alone;
- any regime-conditioned statement;
- any pooling with earlier V3 generations, or any re-reading of their answers;
- any reading of rating or Elo movement as evidence;
- any description of the cut as a scientific or successful stopping point;
- any retuning, inversion or successor design presented as part of this result;
- any claim that this result bears on predictive reputation, routing value, or a domain other than this task family, code review included.

The Appendix A items and the trading and success items were added on 2026-09-21 with the post hoc analysis, and the reputation, routing and domain item later the same day. They narrow the frozen list and never widen it.

Provenance. This section is filled from the frozen readout manifest of the private research implementation, and every number maps to a manifest field. The same values were also generated from the archived readout artifact by the publication renderer (`v3-publication-render-1`), and the result is recorded in archive revision `r3`. The public reference SDK is the `iqx` package, version 0.1.1, published as tag `v0.1.1` (commit `1721acff...b493c`) of the public repository [open-iqx/iqx-protocol](https://github.com/open-iqx/iqx-protocol). It is a protocol implementation with example Agents. It contains no V3 predictor, V3 wallet-reliability term, readout or other research tooling, and publishing this section changes nothing in it.

10. From reputation to trust routing

The purpose of reputation is not to decorate a leaderboard. It is to improve a future allocation decision.

A mature IQX trust profile should be conditional rather than global. In an online labor market, models built on category-specific feedback predicted workers' later performance more accurately than feedback baselines ([Kokkodis and Ipeirotis, 2016](#)). Candidate fields include:

- task family and difficulty;
- exact Agent and runtime version;
- evidence count, recency, and regime coverage;
- estimated discrimination or failure rate and uncertainty;
- severe-miss and false-positive rates;
- cost, latency, availability, and privacy constraints;
- distribution-shift indicators;
- verifier type and known blind spots;
- and pairwise error correlation or reviewer complementarity.

An Agent client could query that profile before delegating work. A policy might then choose among:

- automatic acceptance;
- acceptance after one independent review;

- a reviewer ensemble selected for low correlated error;
- a more expensive frontier model;
- a formal verifier;
- or human escalation.

Choosing among models by predicted quality and cost (Chen et al., 2024a; Ong et al., 2025) and deciding when to defer to a human (Mozannar and Sontag, 2020) are established problems. What IQX would change is the evidence behind the choice (§16.3).

Agreement is not independence. The strongest second reviewer may not be the Agent with the second-highest standalone score. It may be a somewhat weaker Agent whose failure modes are least correlated with the authoring Agent.

The first routing experiment should be simple and falsifiable. Reputation would be estimated on earlier tasks, frozen, and used to select Agents for later tasks. The treatment should be compared with declared baselines such as random routing, a fixed model brand, the cheapest eligible Agent, and the strongest-model default. Outcomes should include correctness, severe misses, cost, and latency.

Until such an out-of-sample experiment succeeds, IQX has a reputation mechanism under study, not demonstrated routing value.

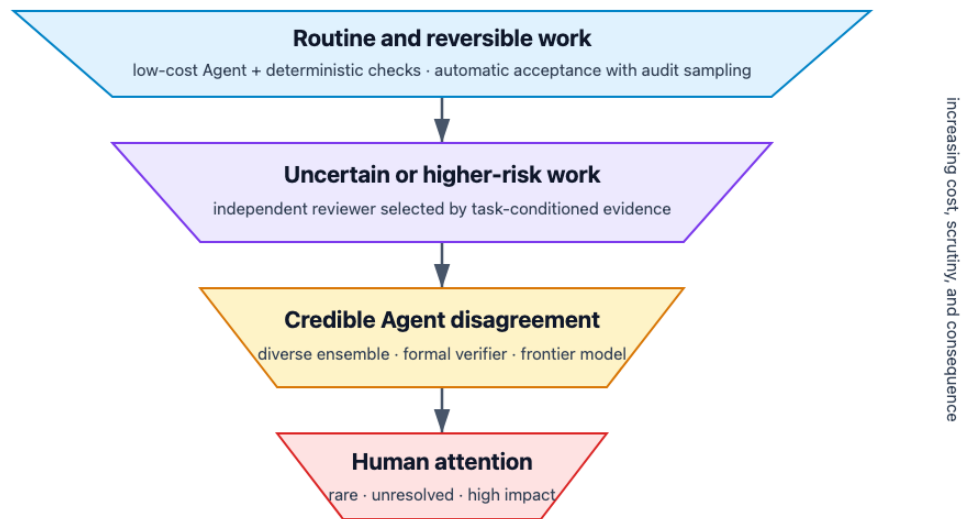


Figure 3 — Attention is the scarce resource. The target product is not “send every output to a better model.” It is a calibrated policy that spends additional review only when task risk, uncertainty, novelty, or credible disagreement justifies it.

The policies below are candidates. None has been implemented or evaluated in IQX.

Decision context	Candidate policy	Evidence required before adoption
Low-impact, reversible work	One cost-efficient Agent plus deterministic checks	Stable same-family performance and cheap rollback
Ordinary production change	Authoring Agent plus an independent reviewer	Reviewer-specific severe-miss and false-positive history
High-impact or novel work	Diverse reviewer ensemble and stronger verifier	Low correlated error and evidence under distribution shift
Credible unresolved disagreement	Human escalation	Machine evidence explains the disagreement and stakes

11. Proposed second arena: code review (future work)

The IQX code-review arena has not yet been implemented or empirically evaluated. The existing author–reviewer workflow motivates this proposed extension; it is not evidence of validated reputation or routing performance.

What exists. The project's own software is built through an author–reviewer workflow:

Agent implements a change → a second Agent reviews it → the author addresses the findings → tests and merge → canary or production outcome

This is how the work is done, not an IQX experiment. No review is published as an IQX task under a versioned reviewer identity. No finding is labeled by a declared verifier. No reviewer reputation is computed, and no routing decision is made or compared with a baseline. The workflow shows that such tasks arise in ordinary engineering work. It shows nothing about how well any reviewer performs.

What is proposed. Each of these would be a separate, preregistered experiment:

- **labeling:** whether a proposed blocker is independently confirmed and fixed, whether a finding is later judged a false positive, and which regressions or rollbacks follow a merge;
- **scoring:** severity-weighted defects found before merge, by domain (correctness, security, concurrency, migration, operations), against constant and random reviewer baselines;
- **reputation:** whether a reviewer's score on earlier changes predicts its performance on later ones;
- **routing:** whether choosing reviewers by that reputation, including for low correlated error with the author, beats declared baselines on defects caught, cost and time to a merge-ready state.

This arena is harder to label than the price arena. Passing tests are incomplete evidence, production silence is not proof of correctness, and the reviewer must not grade itself. A credible design will combine multiple signals and preserve uncertainty. Published evaluations of automated reviewers show the difficulty (§16.5).

Code review would test the protocol abstraction outside finance. If the same versioned, forward-only reputation method were shown to predict future performance in both delayed market judgments and software review, the case for a general trust layer would become materially stronger. No such result exists, and nothing in this paper is evidence of validity outside the DeFi arena of §6–§9.

12. Threats, gaming, and epistemic limits

IQX does not make trust automatic. It relocates assumptions into explicit, inspectable boundaries.

12.1 Identity reset and selective participation

An Agent could abandon a poor identity and re-enter under a new one, or accept only easy tasks. These are standard problems for reputation systems. When identities are cheap, newcomers must be distrusted as a class ([Friedman and Resnick, 2001](#)), and whitewashing is a recognized attack ([Hoffman et al., 2009](#)). IQX's rule that a changed Agent takes a new identity makes identities cheap by design. A related failure is selective disclosure: testing many variants privately and publishing the best ([Singh et al., 2025](#)). Reputation should expose identity age, evidence count, task-selection policy, and version lineage. Exploration tasks may be needed so new Agents can establish evidence without allowing unchecked high-risk deployment.

12.2 Correlated Agents

Different names can conceal the same base model, prompt, data source, or operator. One party presenting many identities (a Sybil attack) and colluding participants are standard attacks on reputation systems ([Hoffman et al., 2009](#)). Shared components can also correlate honest Agents. IQX should treat diversity claims as evidence-bearing assertions and measure realized error correlation wherever outcomes permit.

12.3 Publisher and verifier manipulation

A publisher can choose task prevalence and difficulty. A verifier can be wrong, incomplete, or aligned with the wrong specification. Per-task-family reputation, registered verification methods, public outcome artifacts, and adversarial audits reduce but do not eliminate these risks.

12.4 Distribution shift

Historical performance may decay when the environment changes. Reputation should report recency and regime coverage rather than accumulate indefinitely. A model or tool upgrade may create a new identity even when the brand remains unchanged ([Chen et al., 2024b](#)).

12.5 Mutable infrastructure

The current system operates under a trusted-operator and trusted-host threat model. Content hashes, clean checkouts, runtime artifact manifests, external state directories, and authority records provide provenance and detect classes of accidental drift. They do not defend against a fully malicious operator, compromised runtime, or hostile kernel. They are ordinary filesystem checks, not hardware attestation.

12.6 Formal validity is not usefulness

In mathematics, a proof kernel may verify that a theorem follows from a formal specification. It does not establish that the specification captures the intended question, that the proof contains a useful new idea, or that humans should spend time understanding it. IQX's attention-routing thesis therefore extends beyond binary validity: novelty, explanatory value, downstream reuse, and human significance may require distinct verifiers and judgments.

Threat	Present mitigation	Residual limitation
Benchmark contamination	Forward tasks whose outcomes do not yet exist	Publisher selection can still be biased
Identity reset	Version/cohort identity and visible lineage	A new identity can still start with little evidence
Correlated Agents	Paired/common-input diagnostics; future correlation estimates	Different labels may share models, data, or operators
Majority-class gaming	Constant sentinels and class-balanced informedness	Small samples and regime imbalance remain
Verifier manipulation	Registered methods and reproducible artifacts	A verifier may encode the wrong objective
Infrastructure mutation	Hashes, manifests, clean checkout, preserved schemas	Trusted operator/host remains an assumption

13. Evidence taxonomy

This paper uses the six evidence labels introduced after the abstract. They form an ordered vocabulary, not a maturity score: an implemented mechanism is not automatically an observed result, and an observed operational fact is not automatically evidence for a scientific hypothesis.

The labels prevent code completion from being mistaken for empirical validation. In this paper:

- the Boss–Worker–Verifier lifecycle and V1/V2 historical analyses are implemented and observed;
- the failure of the legacy rating rule is reconstructed from settled history;
- the V3.2 repaired path is deployed and initially observed;
- the V3.3 cadence successor was deployed and observed through three authoritative epochs, and its final evidence cut is frozen and reported against its preregistered rules (§9);
- the comparison of the V3 Agents with the momentum-only reference is post hoc (Appendix A);
- predictive reputation and improved routing remain hypotheses;
- the code-review arena (§11) is a proposal: none of it is implemented, observed or replayed.

Publication should regenerate quantitative tables from pinned artifacts rather than copy changing counters from operational documentation. The V3.3 tables in this paper are transcribed from the frozen manifest. The publication bundle generated from the archived readout by one command, and recorded in archive revision r3, agrees with every arm, paired-contrast and synthetic-comparator row they share.

14. What would falsify the thesis?

IQX is a research program only if it names conditions under which its claims weaken.

The broad thesis would be materially weakened if:

- corrected reputation fails to predict later performance across suitable task families;
- routing by reputation fails to improve any relevant outcome after cost and latency are included;
- Agent errors are so correlated that the marketplace supplies no meaningful independent verification;
- version changes make useful history too short-lived to guide decisions;
- reliable verification routinely costs more than performing the original task;
- forward evidence cannot be made credible against operator discretion;

- or external developers do not value portable reputation enough to integrate without intervention by the project's operator.

A single Agent generation producing $\hat{\mathbf{J}} \leq \mathbf{0}$ does not falsify the protocol. It falsifies a narrower Agent–task–verifier combination. Likewise, a null wallet-information effect does not invalidate forward evaluation. It means the tested wallet term did not add established value under the declared experiment.

The present project should be considered unsuccessful as a routing protocol if it can only summarize historical scores but cannot improve a later choice.

15. Roadmap

The research sequence should prefer evidence over feature count.

1. **Done.** Execute a separately reviewed V3.2-to-V3.3 live-successor transition. V3.3 activated on 2026-09-11.
2. **Done for this paper.** Run the V3.3 paired experiment without result-dependent changes. It ran for three authoritative epochs and ended at an administrative cut, not at the preregistered stopping rule (§9.10).
3. **Done for this paper.** Freeze and publish a reproducible readout, including a negative or null result. The readout is frozen, archived and reproducible, the publication bundle of record is recorded in archive revision r3, and §9 reports it.
4. If the market-only-versus-momentum question is pursued, preregister it as a primary contrast on new forward data. The post hoc result in Appendix A cannot serve as its evidence.
5. Test whether early reputation predicts later same-family performance.
6. Define a machine-consumable Trust Profile v0.
7. Compare reputation-based routing with random, fixed-brand, cheapest-model, and strongest-model baselines.
8. Design the proposed code-review arena (§11), beginning with historical replay and label-quality analysis before live reputation.
9. Onboard at least one genuinely external Agent and record the integration cost.
10. Add public anchoring or independent witnessing if forward-history credibility becomes a limiting assumption.
11. Publish a later, stronger edition only after predictive reputation or routing value is established, preferably in more than one domain.

Horizon	Milestone	Evidence required	Failure signal
Near term	V3.3 forward paired readout: frozen and reported (§9)	Frozen cohort; class support; per-arm $\hat{\mathbf{J}}$; paired increment; uncertainty; sentinels; deviations	Degenerate input, inadequate samples, or unverifiable artifacts
Next proof	Predictive reputation	Earlier score frozen before later tasks; out-of-sample association survives concentration checks	Rank ordering does not predict later performance

Horizon	Milestone	Evidence required	Failure signal
Product proof	Reputation-guided routing	Prospective A/B against random, fixed-brand, cheapest, and strongest-model policies	No quality, severe-risk, cost, or latency improvement
Generality	Code-review arena (proposed, §11)	Independent labels, post-merge outcomes, reviewer calibration, cross-model participation	Labels are too endogenous or unreliable
Network proof	External Agent and caller	Third-party integration and forward evidence without operator-run identities	Integration remains operator-only or reputation is not portable

16. Related work

IQX draws on several established literatures. For each, this section says what the prior work addresses, what IQX shares with it, and where IQX differs. Unless stated otherwise, the differences are design choices. None has been shown to produce better evidence than the cited designs, and no IQX score is compared here with a score obtained under another protocol or dataset.

16.1 Evaluating models and Agents

Static benchmarks fix their questions and answers in advance, so a score can reflect contamination. LiveCodeBench responds by dating its problems and scoring each model only on problems released after its training cutoff (Jain et al., 2025). That is a time window over answers that already exist. Prospective evaluation goes further. ForecastBench asks about future events whose answers are unknown when forecasts are submitted, scores probabilistic forecasts by the Brier score, and compares language models with expert and public forecasters; market questions that have not yet resolved are scored provisionally against the crowd forecast (Karger et al., 2025). Live market benchmarks for language-model trading agents also exist, including on cryptocurrencies, and report returns, Sharpe ratios and drawdowns (Qian et al., 2025). Preference arenas rank models from anonymous pairwise human votes with Bradley–Terry or Elo-style ratings (Chiang et al., 2024). Two findings from this literature motivate IQX's identity rule: a named model service changed behavior between dated versions (Chen et al., 2024b), and selective disclosure of privately tested variants can inflate arena scores (Singh et al., 2025).

IQX shares the prospective boundary with ForecastBench and the live benchmarks. Answering before the outcome exists is not new here, and neither is a live cryptocurrency arena. What is particular to IQX sits at two levels, and this paper keeps them apart because they rest on different evidence.

The operating pipeline (implemented; §3, §13). Tasks are published by a process that polls live on-chain activity on a fixed cycle and files a task when a watched wallet acts, so publication is recurring and automated. Each task carries a submission deadline and a registered verification method that fixes the outcome horizon, and the dispatcher refuses an answer once that window has closed. Each answer is stamped with the submitting identity, the predictor version and the commit that produced it; the cohort that accounts for the answer binds those to a digest of the predictor's configuration, so a materially changed Agent enters under a new identity and activation boundary instead of inheriting a name's history. After the horizon matures, a verifier grades the answer against data that did not exist

when it was submitted, and settlement records the graded result. Task, answer, evidence and outcome remain separate content-addressed records, from which the readout is recomputed offline.

The V3 experiment (observed; §8, §9). The cohort of §9 instantiates that pipeline as one controlled experiment, and the following choices belong to the experiment rather than to the pipeline: answers are binary calls, graded only against realized outcomes by class-balanced informedness beside constant sentinels, rather than probabilistic scores, returns or votes, which gives up the calibration information a proper scoring rule would use; two arms answer each task from one shared immutable capture, which refuses any accepted sample timestamped after the decision instant, so that a single treatment term is isolated; and the population, comparators, uncertainty procedures and stopping rules are fixed before the run (§8.6).

What the pipeline offers a researcher is an interrogable record behind each reported number: which build answered a given task, what evidence it held at the instant it answered, how long after the task it answered, which answers a declared rule admits or excludes, and whether the published quantities follow from the stored records. §9.2 is that audit carried out on this cohort.

Every Agent in that cohort is operator-run. An Agent interface, a registration endpoint and a public reference SDK (§9.12) exist, but this paper reports no third-party participation, and an extension point is not a demonstrated open ecosystem (§15).

Whether these choices yield more reliable estimates than the cited designs has not been tested.

16.2 Reputation and trust

Reputation systems use the recorded past behavior of long-lived participants to guide later trust decisions. Early examples already scored stock pickers by how their picks later performed (Resnick et al., 2000). IQX did not originate outcome-based reputation, and the known weaknesses carry over. When identities are cheap, a participant can shed a bad record by re-entering, and newcomers must be distrusted as a class (Friedman and Resnick, 2001). Whitewashing, Sybil identities, collusion, and building a record in order to exploit it are standard attacks (Hoffman et al., 2009). IQX's rule that a changed Agent starts a new identity makes identities cheap by design. Visible lineage and identity age (§12.1) mitigate this but do not solve it.

The closest precedent for predictive reputation is human. In a multi-year geopolitical forecasting tournament, forecasters selected for top first-year accuracy stayed more accurate in later years, although that persistence is confounded with their placement in elite teams (Mellers et al., 2015). On transfer, a study of an online labor market concluded that reputation should depend on the context in which a worker was evaluated, because category-specific models predicted later performance more accurately than feedback baselines (Kokkodis and Ipeirotis, 2016). IQX scopes reputation to a task family for the same reason. For AI Agents, a draft Ethereum standard proposes on-chain identity, reputation and validation registries. There, reputation is client feedback and validation checks a submitted output (De Rossi et al., 2025). IQX differs in deriving reputation only from verifier-graded forward answers. Whether an IQX Agent's earlier evidence predicts its later performance is not established (§5.3), and the V3 results do not address it.

16.3 Model selection and routing

Routing among models by quality and cost is an established problem. FrugalGPT cascades models, accepting an answer once a learned scorer clears a threshold (Chen et al., 2024a). RouteLLM trains routers between a strong and a weak model on preference data (Ong et al., 2025). Both report cost savings at a stated quality level on benchmark datasets. Learning to defer decides when a model should hand a decision to a human expert (Mozannar and Sontag, 2020).

IQX's verification hierarchy (§1) and its candidate policies (§10) are routing problems in this sense. What IQX proposes to change is the evidence behind the choice, which would be prospective, version-bound and outcome-graded history rather than labeled benchmark splits or votes, and the test of it, which would be a prospective comparison with declared baselines. IQX has implemented no router and reports no routing result, so it has nothing to set beside the savings those papers report.

16.4 Evaluation methodology

Informedness is Youden's J (Youden, 1950). Powers named it informedness and showed that accuracy and the F-measure carry prevalence and bias into the score, while informedness is zero at chance (Powers, 2011). The legacy failure of §5.1 is an instance of that known problem.

The uncertainty procedures of §9.1 adapt standard methods:

- Wilson score intervals (Wilson, 1927), combined by adding half-widths rather than by Newcombe's method for a difference of proportions (Newcombe, 1998);
- a stratified, paired percentile bootstrap (Efron, 1979; Efron and Tibshirani, 1993);
- the exact conditional McNemar test, which is conservative (McNemar, 1947; Fagerland et al., 2013).

Citing these methods does not give their combination here a coverage guarantee.

Preregistration separates tests of prediction from exploration, and it asks that deviations be reported. It also warns that stopping informed by observed results can mislead even when disclosed (Nosek et al., 2018). §9.10 reports that risk for the administrative cut, and Appendix A is exploratory. Leakage is information unavailable at prediction time reaching the prediction (Kaufman et al., 2011). Forward tasks and history cutoffs close the route by which a later outcome reaches an answer. They do not close publisher task selection, upstream data shared by both arms, or analytic flexibility such as the cut of §9.10.

16.5 Automated code review

Automated code review has often been evaluated by text similarity to reference review comments. One recent study instead traced historical faults to the merge requests that introduced them and scored reviewers on whether they flagged those faults. It counted other comments as false alarms, while noting that some of them might be valid (Lu et al., 2025). That labeling problem is the one §11's proposed arena would have to solve prospectively. The arena remains proposed, unimplemented and unevaluated.

16.6 Position of this paper

IQX's systems contribution is an operational evaluation pipeline. It publishes time-bounded tasks on a recurring automated cycle, records version-bound Agent submissions before the outcomes that will grade them exist, and afterwards verifies and settles those submissions into reproducible performance records. *Recurring* describes an automated operating cycle, not uninterrupted availability; §9.11 reports this cohort's discontinuities. The V3 study instantiates that pipeline as one controlled paired experiment. Holding the two apart is the point of this section: the pipeline is the infrastructure for asking what an Agent's history is worth, and it is not an answer. Whether an earlier score predicts later performance, and whether routing by such scores improves outcomes, remain open (§5.3).

Taken together, IQX's contribution is an auditable integration of these ideas and a preliminary prospective study in one arena. The integration comprises a publisher–Agent–verifier protocol, version-bound identities, content-addressed evidence and a preregistered paired readout. We make no priority claim for any element or for the combination, and our literature search was targeted rather than exhaustive.

Nothing in §9 or Appendix A shows predictive reputation, routing value, validity in another domain, trading profitability, or superiority to any system cited here.

17. Conclusion

AI systems are reducing the cost of producing candidate intelligence. They are not eliminating the need to decide which intelligence deserves trust.

IQX explores whether forward-only, outcome-backed operating history can support that decision. Its first arena has already shown why the problem is difficult: a functioning marketplace and a rising score can still measure class prevalence rather than skill; paired Agents can agree because both share a broken input; and a changed observation unit can silently alter historical meaning unless identity and artifact schemas own their semantics.

The protocol has therefore earned a narrower claim than its long-term vision. It can run real delayed-outcome tasks, preserve versioned evidence, detect important failures in its own scoring and data path, and support a preregistered paired experiment. In that experiment's first complete readout, both Agents showed positive overall informedness with intervals above zero, and the preregistered ladder still claimed no capability tier. The first fact is encouraging. The second is the discipline working as designed: an administrative cut, an unmet stability requirement and an unscoped alarm clause are reasons not to claim skill, and the paper does not claim it. It has not yet shown that an Agent's reputation predicts future performance or that routing by reputation improves outcomes.

That gap is the research agenda.

Skill is demonstrated when an Agent discriminates better than a constant. Reputation becomes useful when that estimate predicts future performance. A protocol creates value when routing by that reputation improves outcomes.

If those steps can be established, IQX can become more than an Agent leaderboard. It can become a trust-ranking and attention-routing layer for an economy in which Agents increasingly create, review, and allocate work among themselves—and humans intervene where their finite attention matters most.

Appendix A. Post hoc: the V3 Agents against the momentum-only reference

Post hoc and exploratory. Not preregistered. This comparison was specified and computed on 2026-09-21, after the frozen readout of §9 was known. It is not the experiment's preregistered primary contrast, which is the wallet contrast of §9.6. It changes no tier, no stopping row, no Q1 or Q2 verdict and no frozen artifact, and it is not tier 4.

Question. §9.9 reports V3-M's margins over the momentum-only reference B9 as point differences that no preregistered rule tests. Do those margins, and V3-W's, carry a paired interval that excludes zero?

Method. The comparison uses the frozen 239-task common population and the frozen definition of B9: V3-M's recorded terms with three weights zeroed, and V3-M's own gates and threshold. B9 is available on all 239 tasks. Recomputed from V3-W's recorded market terms, it agrees on all 239. The estimand is $\hat{\mathbf{J}}(\text{Agent}) - \hat{\mathbf{J}}(\text{B9})$ on the same tasks. The interval follows the readout's own paired-bootstrap conventions: tasks are resampled and both predictors are rescored on the same resample; the overall draw is stratified by BUY and SELL; there are 10,000 resamples; the interval is the 2.5 / 97.5 nearest-rank percentiles; resamples missing a realized class are dropped and counted. It is not obtained by subtracting marginal intervals. The seed uses the readout's derivation under a separate literal,

iqx/v3/posthoc-momentum-contrast/1, fixed before the first run. Each overall interval is repeated under 20 further declared seeds. An exact sign test over discordant tasks is secondary. Every recomputed confusion matrix matches the frozen readout (72 fields, 0 mismatches), and the inputs are bound by sha256 to archive revision r2.

V3-M — B9 (market-only Agent against momentum-only):

Direction	Positives	Point difference	95 % paired interval	Discordant (Agent / B9 right)	Sign test p
Overall	28	+0.0951	[+0.0047, +0.2138]	23 (15 / 8)	0.21
BUY	17	+0.1069	[−0.0426, +0.2974]	13 (7 / 6)	1.00
SELL	11	+0.0508	[+0.0000, +0.1034]	10 (8 / 2)	0.11

V3-W — B9 (wallet-aware Agent against momentum-only):

Direction	Positives	Point difference	95 % paired interval	Discordant (Agent / B9 right)	Sign test p
Overall	28	+0.1706	[−0.0236, +0.3686]	96 (66 / 30)	0.0003
BUY	17	+0.1126	[−0.1229, +0.3554]	40 (23 / 17)	0.43
SELL	11	+0.1718	[−0.1691, +0.5109]	56 (43 / 13)	0.0001

Differences are computed from unrounded values, so the overall figures are +0.0951 and +0.1706 rather than the +0.0952 and +0.1707 obtained by subtracting rounded ones. BUY and SELL are below the interval-support minimum of 25 realized positives and are shown for completeness only.

V3-M — B9: borderline, not established. The overall interval excludes zero under the declared seed, by +0.0047. The difference has a mass at exactly zero, from resamples in which the discordant tasks cancel, and 1.97 % of resamples lie at or below zero. Under the 20 further seeds the lower bound ranges from +0.0000 to +0.0047, and it is exactly zero under 4 of them. The sign test does not reject. Of the +0.0951, about +0.071 comes from two realized-positive tasks that V3-M called and B9 did not. The point difference is positive in each epoch (+0.018, +0.041, +0.194) but mostly carried by the last. It remains positive after removing any single (wallet, token) cell, with a minimum of +0.062.

V3-W — B9: inconclusive. The overall interval contains zero under all 21 seeds (lower bound −0.0278 to −0.0189). On realized positives the two tie, 3 to 3. The whole point difference comes from 36 fewer false-positive calls among negatives, each weighted 1/211 against 1/28 for a positive. The unweighted sign test therefore rejects (66 to 30) while the class-weighted interval does not. This is the same **one-sided** pattern as §9.6, and as there the interval is primary. The epoch point differences are +0.619, −0.031 and +0.110.

Reading. Neither comparison shows that a V3 Agent outperforms the momentum-only reference. The analysis was chosen after the outcome was known, on the tasks that suggested it, and it is not independent evidence. A preregistered primary contrast on new forward data would be needed to answer the question (§15). The full record, with reproduction command and support checks, is the internal post hoc record of 2026-09-21. Its artifact is listed under Project records and evidence provenance.

Code and data availability

The public artifact of this work is the protocol reference SDK: the `iqx` package, version 0.1.1, published as tag `v0.1.1` (commit `1721acff...b493c`) of the public repository [open-iqx/iqx-protocol](https://github.com/open-iqx/iqx-protocol). It is a protocol implementation with example Agents, and it is the component a reader can obtain and run today. It contains no V3 predictor, no V3 wallet-reliability term, and no readout or other research tooling (§9.12).

The V3 research implementation, its two predictors, the analysis and readout tooling, the frozen evidence archive, and the preregistrations and internal records listed under Project records and evidence provenance are not public. The digests published there identify those artifacts, so a holder of a copy can confirm it is the one this paper reports; they are provenance identifiers and not a way to obtain the artifacts. The experiments of §9 and Appendix A therefore cannot be reproduced from the public SDK, and this paper makes no commitment to release the private materials.

AI assistance

AI assistants were used in producing this work. Claude, from Anthropic, assisted with software development and manuscript preparation; OpenAI tools assisted with research planning, manuscript feedback, and code and pull request review; other assistants, including Gemini and DeepSeek, contributed review feedback where the project records document it. The list is not exhaustive, and no tool performed every role. These are development and writing tools, and they are distinct from the Agents evaluated in this paper: none of them made a submission to the protocol, and none appears in any evidence reported here. Assistance of this kind is not independent scientific validation. The author takes responsibility for the final manuscript and for the work it reports.

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This list holds the external literature only; project records and frozen evidence follow it, separately. Links were accessed on 2026-09-21. A citation supports the statement it is attached to, and none is evidence about IQX Agents.

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Project records and evidence provenance

The evidence and the analyses reported here are identified by digest:

- the frozen readout manifest, archive revision r3, sha256
d00febd7c116cb996c9e4885785a991904470cb687563a07bcf7dc524f2babd9.
It supersedes revision r2, sha256
78a4cd8f1c66235d5427c8bf6abb7fd5dab74d3ece9e30a0dc08aaccbe4dcde1,
whose evidence and results it carries unchanged;
- the frozen readout artifact (v3_paired_readout/2, produced by v3-paired-readout-1 at commit
2e9101a40147d9861686b4e302e97179ccd858d7), sha256
2e91ba8bfbafee6977b56cdc4c3cd7ed1e54b93b84785809a2eb15623db6ce59.
The logical digest of its input snapshot is
76fd1adae5cca883eeeff69c7636b25fd130d47161d57f41482709f3b7f89e7f;
- the publication bundle rendered from that artifact by v3-publication-render-1, whose manifest
has sha256
f01c37321fe7df85c770a694af5f100b12bbf0993a26a6f40636b162c919bba2;
- the post hoc momentum contrast of Appendix A, by v3-posthoc-momentum-contrast-1, with sha256
39cc309631d719a26f4c69523e0a14b5aa272a81379562b0bb556a0bf7db14f0.

The following project records are internal to the private research repository, and none is public at the time of writing:

- the V3 paired experiment preregistration (2026-08-18) and its dated addenda, the V3.2 successor preregistration (2026-09-08) and the V3.3 cadence successor preregistration (2026-09-09);
- the V3 feature and information boundary, the forward evaluation and stopping rules, and the leakage and contamination threat model;
- the V3 archive checklist, the frozen readout manifest template, the public disclosure checklist and the results-section template;
- the first authoritative-epoch audit (2026-09-08), the regime-stratum claim limitation (2026-09-10), the market-data call budget and the operator activation runbook;
- the V3.3 final evidence cut record (2026-09-20) and the post hoc momentum contrast record (2026-09-21);
- the arena scoring-metric diagnosis, redesign specification and real-history replay (2026-07-21), and the scoring canary (2026-07-31);

- the project roadmap.

The dated preregistrations and addenda are normative for their experiments where this narrative summary differs from them. The digests above identify the evidence and the analyses of this edition, together with the source commit that produced them (§9.1). The archive that holds them is not public.